$$\varepsilon = -\frac{d\Phi}{dt} = -\frac{d}{dt}\left(Bax\right) = Ba\mathrm{v}(t)\,. \tag{1}$$

:

$$\varepsilon = IR\,, \tag{2}$$

:

$$I(t) = \frac{Ba\mathrm{v}(t)}{R}\,. \tag{3}$$

.

:

$$F_A(t) = BaI(t) = \frac{B^2a^2\mathrm{v}(t)}{R}\,. \tag{4}$$

:

$$m\frac{dv(t)}{dt} = F - \frac{B^2a^2\mathrm{v}(t)}{R}\,. \tag{5}$$

(5) $\mathrm{v}(0) = 0$:

$$\mathrm{v}(t) = \frac{FR}{B^2a^2}\left[1 - \exp\left(-\frac{B^2a^2}{mR}t\right)\right]. \tag{6}$$

3. (6) :

$$x(t) = \int_0^t \mathrm{v}(t)dt = \frac{FR}{B^2a^2}\left[t + \frac{mR}{B^2a^2}\left(\exp\left(-\frac{B^2a^2}{mR}t\right) - 1\right)\right]. \tag{7}$$

$$x(t_0) = a\,, \tag{8}$$

: t_0

$$t_0 + \frac{mR}{B^2a^2}\left[\exp\left(-\frac{B^2a^2}{mR}t_0\right) - 1\right] = \frac{B^2a^3}{FR}\,. \tag{9}$$

(9) . t_0 .

(6) $\tau \sim mR/B^2a^2$ $\mathrm{v}_0 = FR/B^2a^2$.

0 τ $w = F/m$,

v_0 :

$$t_0 \sim \tau + \frac{a - \dfrac{wt^2}{2}}{\mathrm{v}_0} = \frac{B^2a^3}{FR} + \frac{mR}{2B^2a^2} = 10.25\,\mathrm{c}. \tag{10}$$

(9) $t_0 \approx 10.5\,\mathrm{c}$.

4. t_0 w .

, :

$$\mathrm{v}(t) = \begin{cases} \dfrac{FR}{B^2a^2}\left[1 - \exp\left(-\dfrac{B^2a^2}{mR}t\right)\right], & t < t_0 \\ \dfrac{FR}{B^2a^2} + \dfrac{F}{m}(t - t_0), & t \geq t_0 \end{cases}. \tag{11}$$

: